

Task 1: House Price Prediction (Linear Regression)

Domain: Real Estate

Goal

Predict the continuous price of a house based on its characteristics.

Part A - Theory

Input: numerical and categorical features (e.g., area, number of rooms)

Model: Multiple Linear Regression

Loss: Mean Squared Error (MSE)

Explain the meaning of regression coefficients and the concept of multicollinearity

Part B - Python

Load a dataset from Kaggle (e.g., California Housing or Ames Housing)

Train Linear Regression model

Report MSE, MAE, and R-squared metrics

Task 2: Heart Disease Classification (Logistic Regression)

Domain: Healthcare

Goal

Classify whether a patient has a high risk of heart disease.

Part A - Theory

Input: patient medical metrics (blood pressure, cholesterol, age)

Model: Logistic Regression

Loss: Log-Loss (Binary Cross-Entropy)

Explain how the sigmoid function converts linear outputs into probabilities

Part B - Python

Load a dataset from Kaggle (e.g., Heart Disease Dataset)

Train Logistic Regression

Plot Confusion Matrix and report Accuracy, Precision, and Recall

Task 3: Customer Churn Prediction (Random Forest Classifier)

Domain: Telecommunications

Goal

Predict if a customer will leave the service (churn).

Part A - Theory

Input: customer activity history, demographics, and billing data

Model: Random Forest

Explain the concept of ensembling (Bagging) and how a single decision tree is split (Entropy or Gini impurity)

Part B - Python

Load a dataset from Kaggle (e.g., Telco Customer Churn)

Train Random Forest Classifier

Plot Feature Importance and report ROC-AUC score

Task 4: Customer Segmentation (K-Means Clustering)

Domain: Retail / Marketing

Goal

Group customers into distinct segments based on purchasing behavior.

Part A - Theory

Input: customer features (e.g., annual income, spending score) without class labels

Model: K-Means

Explain centroid initialization, distance calculation, and the Elbow method for choosing the number of clusters

Part B - Python

Load a dataset from Kaggle (e.g., Mall Customer Segmentation Data)

Scale features using StandardScaler

Train K-Means and plot a 2D scatter plot with colored clusters

Task 5: Binary Flower Classification (Single Neuron)

Domain: Botany

Goal

Classify two distinct species using a single mathematical neuron.

Part A - Theory

Input: physical dimensions (e.g., petal length and width)

Model: Single Artificial Neuron (Perceptron / Adaline)

Explain linear separability and how the step activation function works

Part B - Python

Load the classic Iris dataset (filter to keep only 2 classes for binary classification)

Implement and train a single neuron (using scikit-learn or NumPy)

Plot the decision boundary line

Task 6: Spam Email Detection (Naive Bayes)

Domain: NLP / Cybersecurity

Goal

Classify text messages or emails as spam or not spam.

Part A - Theory

Input: vectorized text (Bag of Words or TF-IDF)

Model: Multinomial Naive Bayes

Explain Bayes' theorem and why the algorithm is called "naive"

Part B - Python

Load a dataset from Kaggle (e.g., SMS Spam Collection)

Convert text to numerical vectors using CountVectorizer or TfidfVectorizer

Train Naive Bayes and report F1-score

Task 7: Bike Sharing Demand (Decision Tree Regressor)

Domain: Urban Mobility

Goal

Predict the number of bikes rented per hour based on weather and time.

Part A - Theory

Input: weather conditions, day of week, time of day

Model: Decision Tree Regressor

Explain the difference between classification trees and regression trees (e.g., using variance reduction to split nodes)

Part B - Python

Load a dataset from Kaggle (e.g., Bike Sharing Demand)

Train Decision Tree Regressor

Investigate the effect of the `max\_depth` hyperparameter on overfitting and report MAE

Task 8: Credit Risk Scoring (K-Nearest Neighbors)

Domain: Finance

Goal

Predict if a loan applicant will default on their loan.

Part A - Theory

Input: credit history, income, loan amount

Model: K-Nearest Neighbors (KNN)

Explain why feature scaling is critical for KNN and how the choice of distance metric affects the model

Part B - Python

Load a dataset from Kaggle (e.g., Credit Risk Dataset)

Apply Min-Max or Standard scaling

Train KNN, experiment with different values of K, and report Accuracy

Task 9: Medical Cost Prediction (Random Forest Regressor)

Domain: Insurance

Goal

Predict individual medical costs billed by health insurance.

Part A - Theory

Input: age, gender, BMI, smoking status, region

Model: Random Forest Regressor

Explain how predictions from multiple decision trees are aggregated to output a continuous value

Part B - Python

Load a dataset from Kaggle (e.g., Medical Cost Personal Datasets)

Apply One-Hot Encoding for categorical variables
Train Random Forest Regressor
Plot "True Values vs. Predicted Values" and report R-squared